section 1

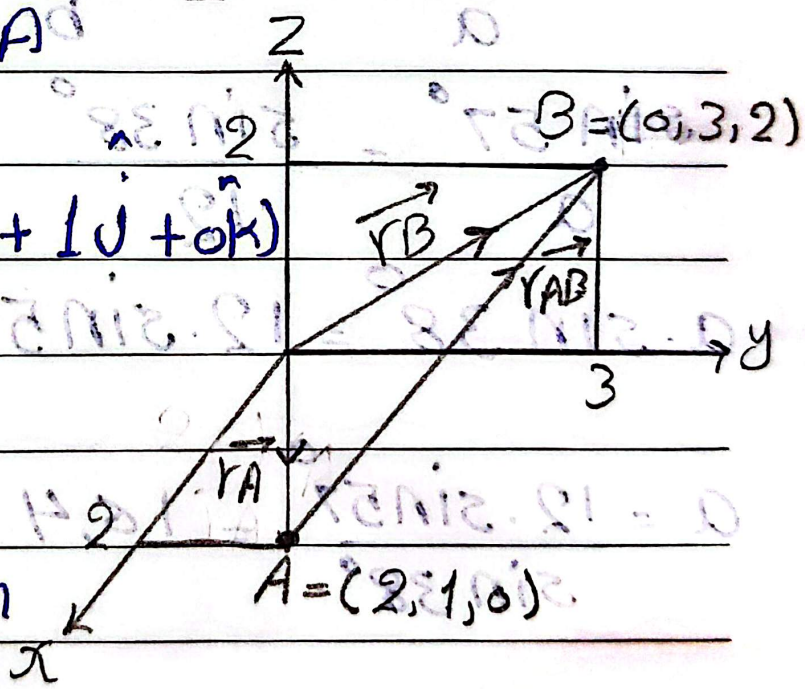
→ let A (2, 1, 0) and B (0, 3, 2) be two Points in the xyz Coordinate system. find the Position vector of Point B with respect to Point A-

↳ $\vec{r}_{AB} = \vec{r}_B - \vec{r}_A$

$= (0\hat{i} + 3\hat{j} + 2\hat{k}) - (2\hat{i} + 1\hat{j} + 0\hat{k})$

$= -2\hat{i} + 2\hat{j} + 2\hat{k}$

$= 2(-\hat{i} + \hat{j} + \hat{k})$ m

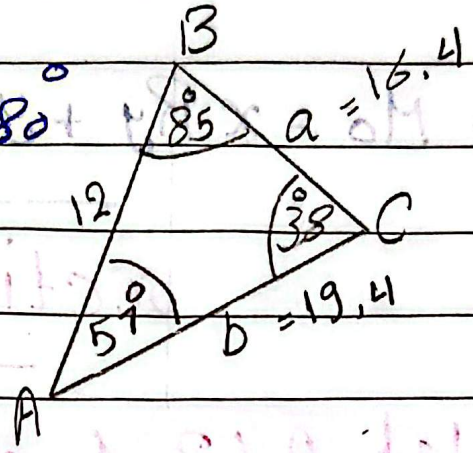


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Solve the following triangle by using the law of sines.



$m\angle A + 85^\circ + 38^\circ = 180^\circ$

$\therefore m\angle A = 57^\circ$

Now, use the law of sines to set up ration For a and b.

$$\frac{\sin 57^\circ}{a} = \frac{\sin 85^\circ}{b} = \frac{\sin 38^\circ}{12}$$

$$\frac{\sin 57^\circ}{a} \times \frac{\sin 38^\circ}{12}$$

$$a \cdot \sin 38^\circ = 12 \cdot \sin 57^\circ$$

$$a = \frac{12 \cdot \sin 57^\circ}{\sin 38^\circ} = 16,4$$

$$\frac{\sin 85^\circ}{b} \times \frac{\sin 38^\circ}{12}$$

$$b \cdot \sin 38^\circ = 12 \sin 85^\circ$$

$$b = \frac{12 \cdot \sin 85^\circ}{\sin 38^\circ}$$

$$b = 19,4$$

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Memo No. _____

Date / /

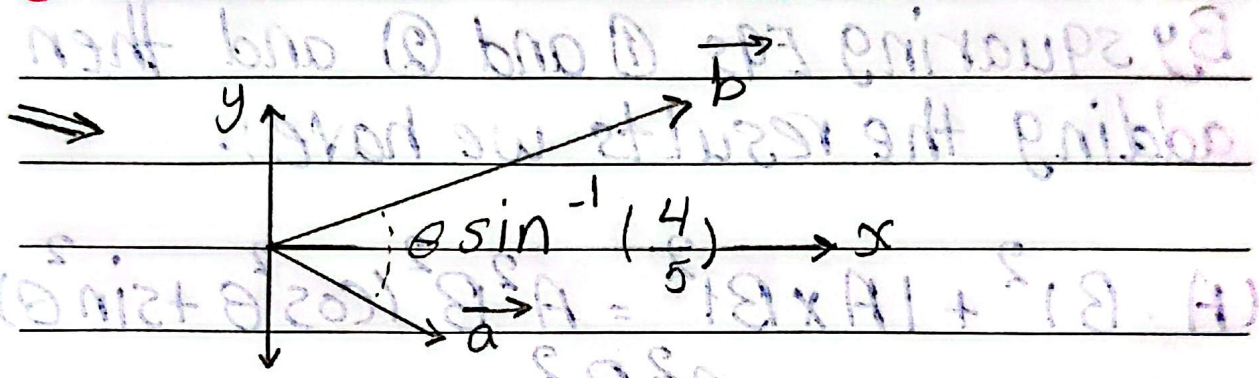
Find the cross Product Given

$\vec{a} = 2\hat{i} - \hat{j}$ $\vec{b} = 4\hat{i} + 2\hat{j}$

Find the cross Product of the two vectors

① using the angle θ

② using the components of the vectors



⇒ Cross Product using the angle θ

$\therefore a \times b = |\vec{a}||\vec{b}| \sin\theta \, n$

$= \sqrt{5}\ \sqrt{20} \sin(\sin^{-1}\frac{4}{5})k = 8\hat{k}$

⇒ Cross Product using the components of the vectors.

$$a \times b = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 0 \\ 4 & 2 & 0 \end{vmatrix} = 8\hat{k}$$

Mo Tu We Th Fr Sa Su

Memo No. _____

Date / /

Prov that $(A \cdot B)^2 = A^2B^2 - |A \times B|^2$

$\Rightarrow$

$$A \cdot B = AB\cos\theta \quad (1)$$

$$A \times B = AB\sin\theta n$$

$$\therefore |A \times B| = AB\sin\theta \quad (2)$$

By squaring Eqs (1) and (2) and then adding the results we have:-

$$(A \cdot B)^2 + |A \times B|^2 = A^2B^2(\cos^2\theta + \sin^2\theta)$$

$$= A^2B^2$$

$$\therefore (A \cdot B)^2 = A^2B^2 - |A \times B|^2$$

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Find the unit vector which perpendicular on both two following vectors

$$\vec{A} = \hat{j} + \hat{k}$$

$$\vec{B} = 6\hat{i} - 3\hat{j} + 4\hat{k}$$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 1 & 1 \\ 6 & -3 & 4 \end{vmatrix} = 7\hat{i} + 6\hat{j} - 6\hat{k}$$

$$\therefore |\vec{A} \times \vec{B}| = \sqrt{7^2 + 6^2 + (-6)^2} = 11$$

$$e = \frac{\vec{A} \times \vec{B}}{|\vec{A} \times \vec{B}|} = \frac{7}{11}\hat{i} + \frac{6}{11}\hat{j} - \frac{6}{11}\hat{k}$$

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